

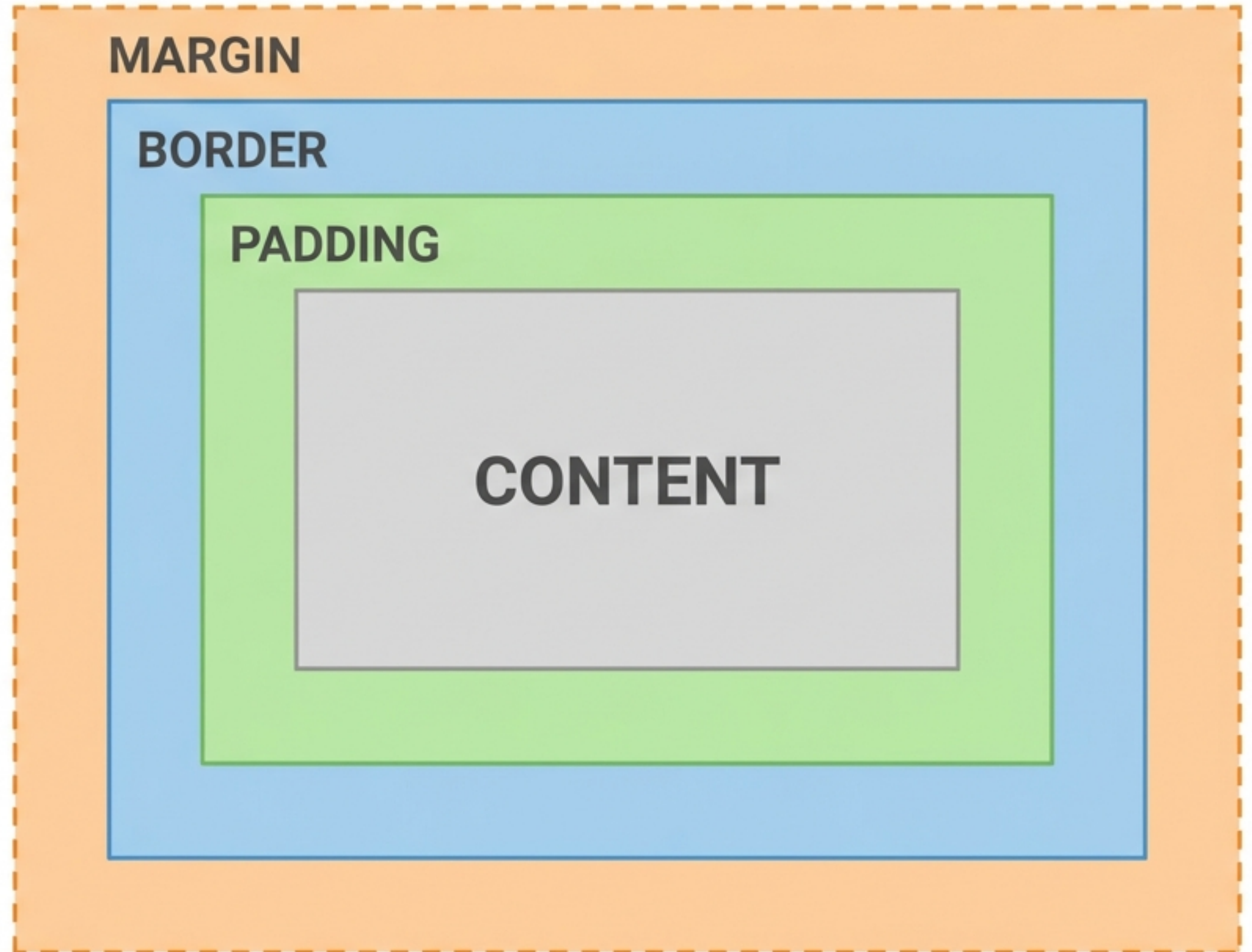
Client-Side Technologies: Intermediate CSS

Day 2 – From Structure to Style

Instructor: Dr. Niveen Nasr El-Den
Institution: iTi

The Box Model Paradigm

Concept: All HTML elements can be considered as boxes. This model allows us to place borders around elements and space them in relation to others.



Spacing the Box: Margins & Padding

Padding (Internal Space)

The area between the content and the border.

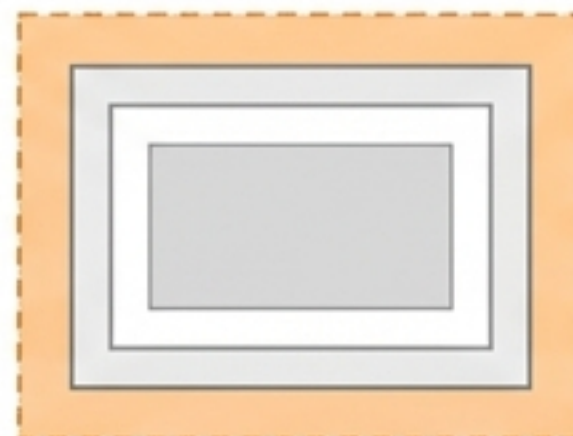
```
1 /* Specific */
2 padding-top: 10px;
3 padding-right: 15px;
4 padding-bottom: 10px;
5 padding-left: 15px;
6
7 /* Shorthand */
8 padding: 10px 15px;
```



Margin (External Space)

The area outside the border, creating personal space.

```
1 /* Specific */
2 margin-top: 20px;
3 margin-right: auto;
4 margin-bottom: 20px;
5 margin-left: auto;
6
7 /* Shorthand */
8 margin: 20px auto;
```



Measurement Units:

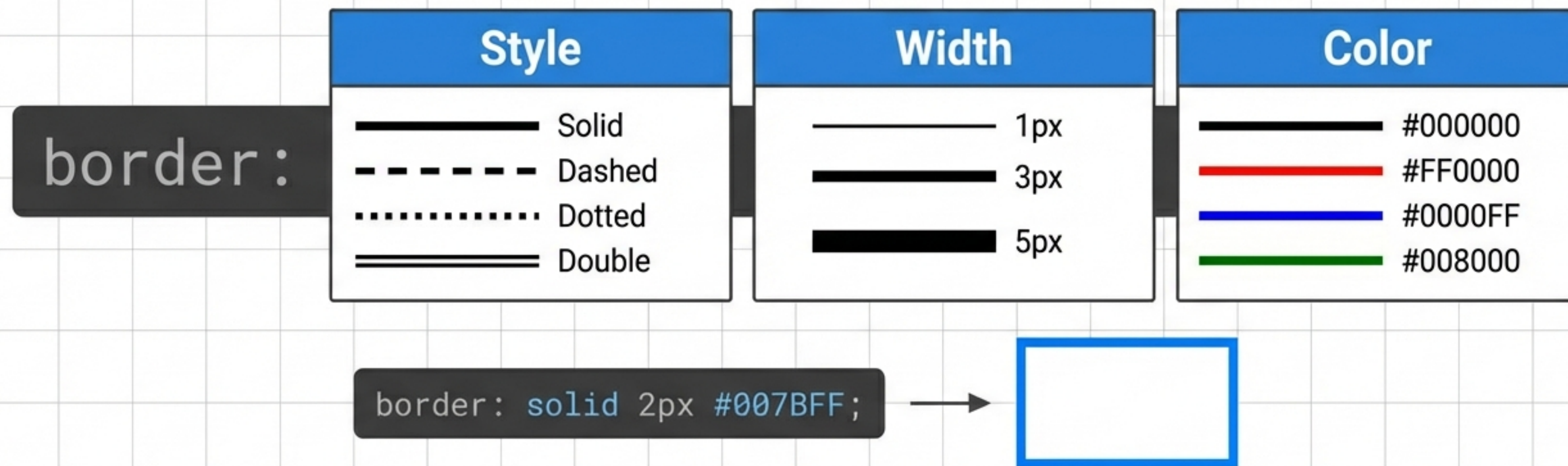
Physical: in, pt
Screen: px

Relative: %, em

Note: Zero values require no unit (e.g., margin: 0;)

Defining the Border

The visible shell of the box requires three properties: **Style**, **Width**, and **Color**.



Measurement Units:

Physical: in, pt
Screen: px

Relative: %, em

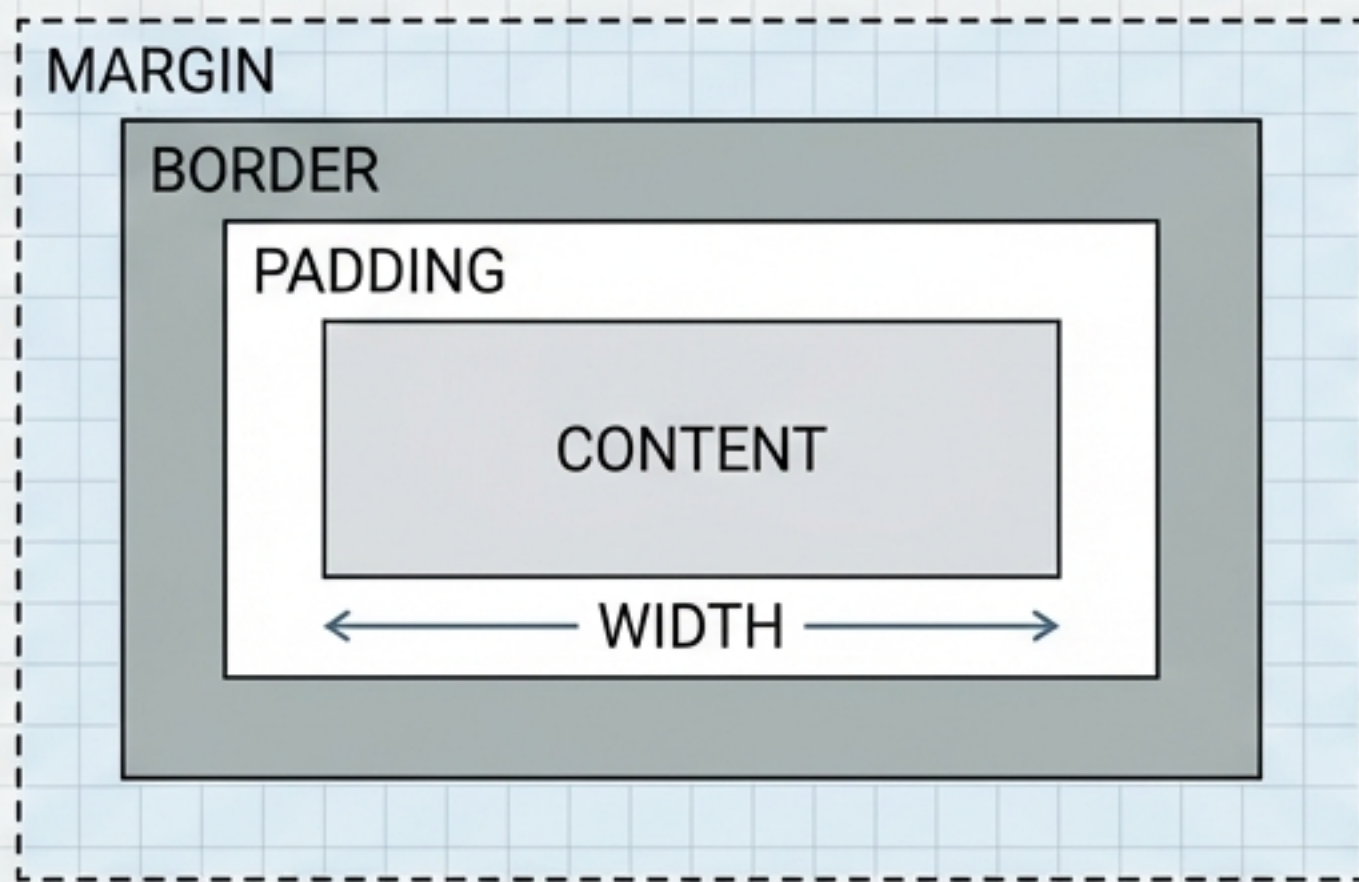
Note: Zero values require no unit (e.g., `border: 0;`)

Rendering Modes: Standards vs. Quirks

Why layouts break: The History Lesson

Standards Mode (W3C)

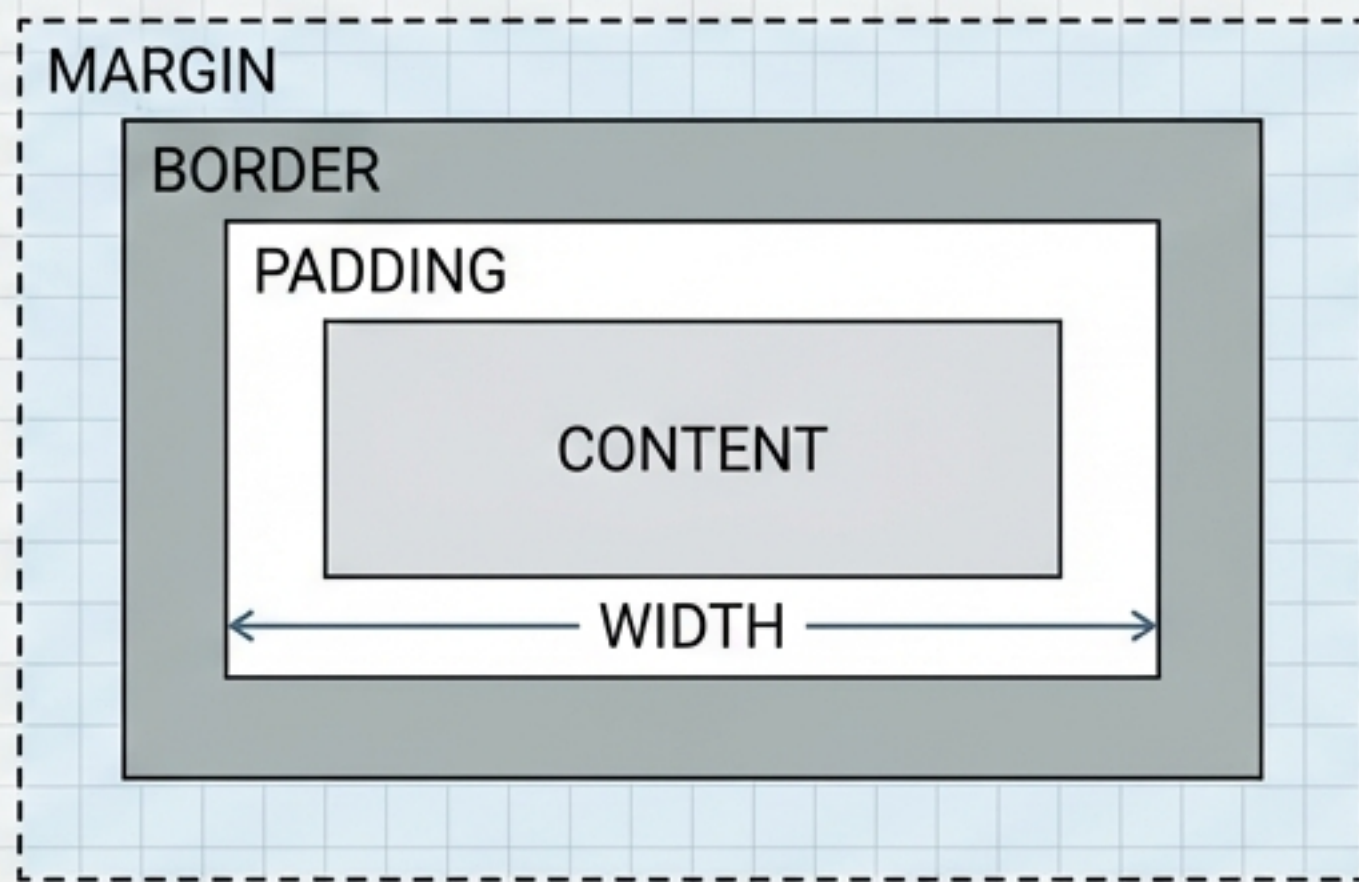
Browser follows HTML/CSS specifications.



Width = Content

Quirks Mode (Old IE)

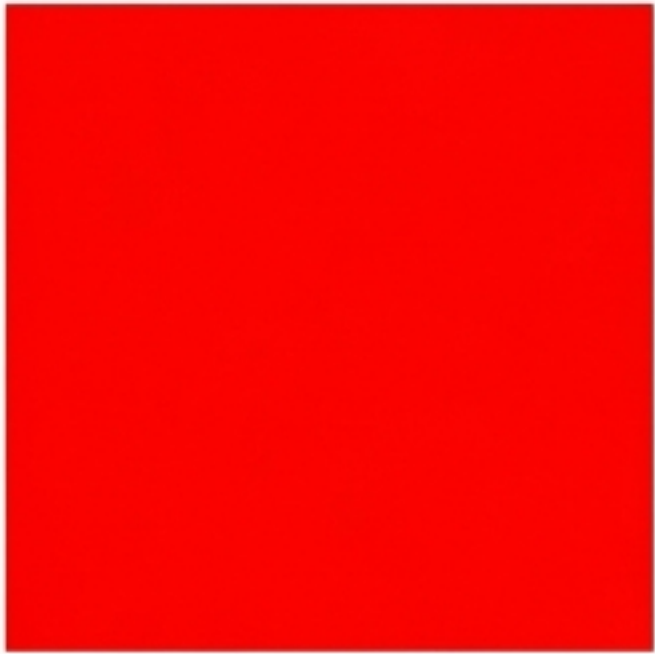
Emulates legacy browsers (IE5) to support old sites.



Width = Content + Padding + Border

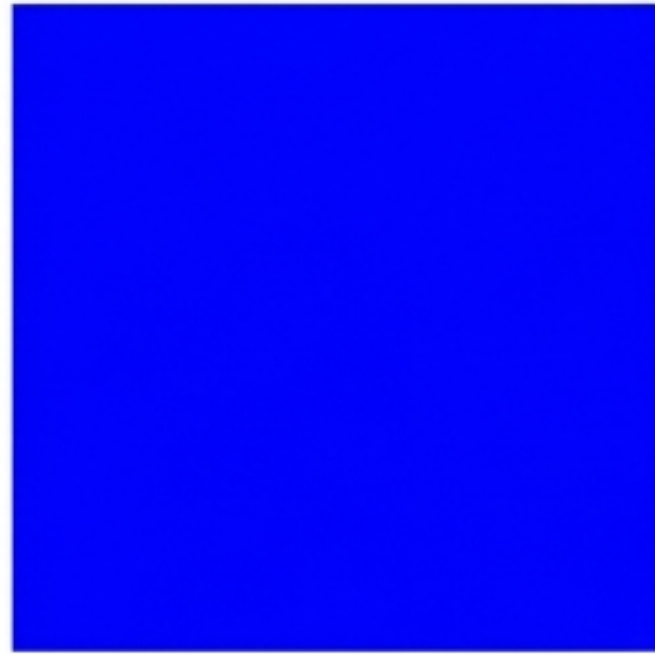
Color Systems & Transparency

Keywords



`color: red;`

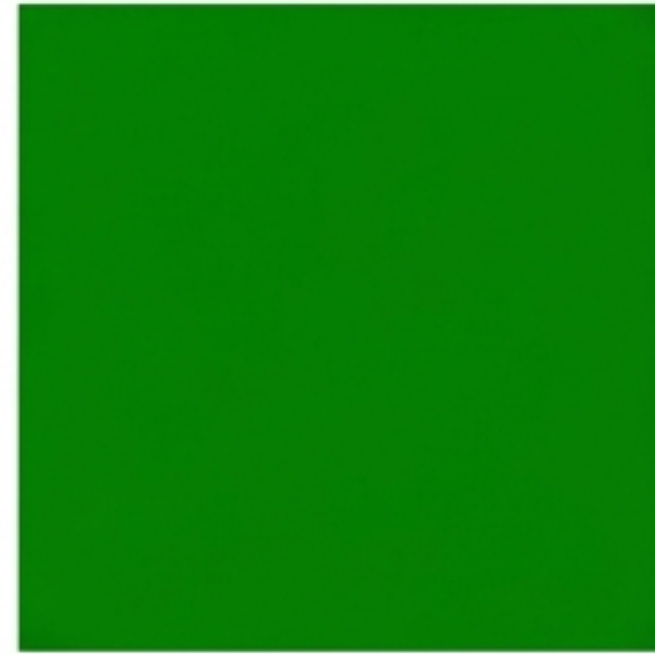
Hexadecimal



`#RRGGBB` or `#RGB`

Example: `#0000FF` or `#00F`

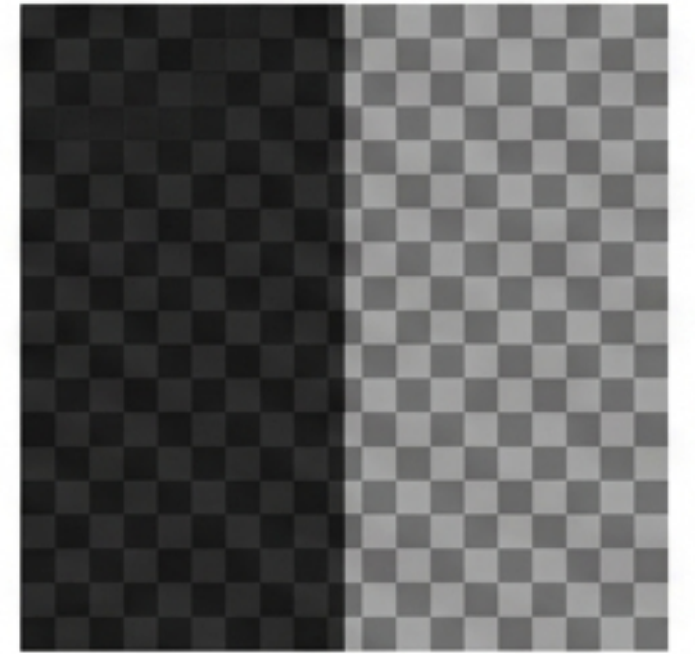
RGB



`rgb(R, G, B)`

Example: `rgb(0, 128, 0)`

RGBA (Alpha)



`rgba(R, G, B, A)`

Example: `rgba(0, 0, 0, 0.5)`

Properties: `color: [value]` (Foreground/Text) `background-color: [value]`

Typography & Text Decoration

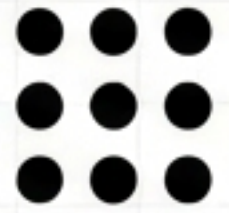
PROPERTY	CSS CODE	VISUAL SPECIMEN
Font Selection	font-family: Arial, sans-serif; font-size: 16px; font-weight: bold;	Arial Bold 16px
Text Alignment	text-align: right;	Aligned Right
Decoration	text-decoration: underline; line-through;	<u>Underline</u> , Deleted
Transformation	text-transform: uppercase; capitalize;	MAKE ME UPPERCASE Capitalize First Letter
Spacing	line-height: 1.5; letter-spacing: 2px;	S p a c e d T e x t

Mastering Background Images

```
background-image: url('image.jpg');
```

Repeat

`background-repeat`



repeat



no-repeat



repeat-x



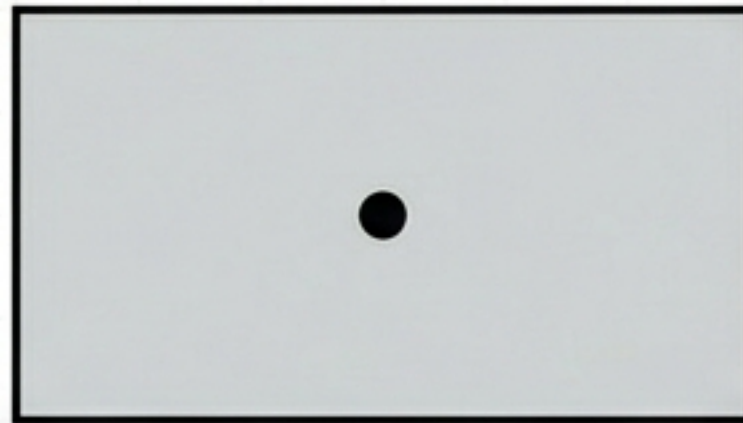
repeat-y

Position

`background-position`

x-axis | y-axis

left, center, right | top, center, bottom



center center

Attachment

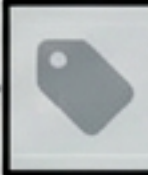
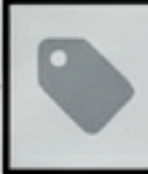
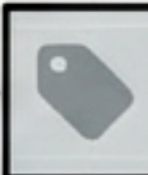
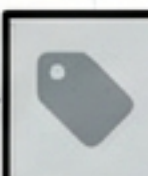
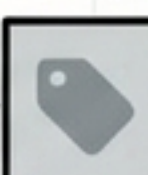
`background-attachment`

scroll (moves with page)

fixed (static window)

Advanced Selection: Attributes

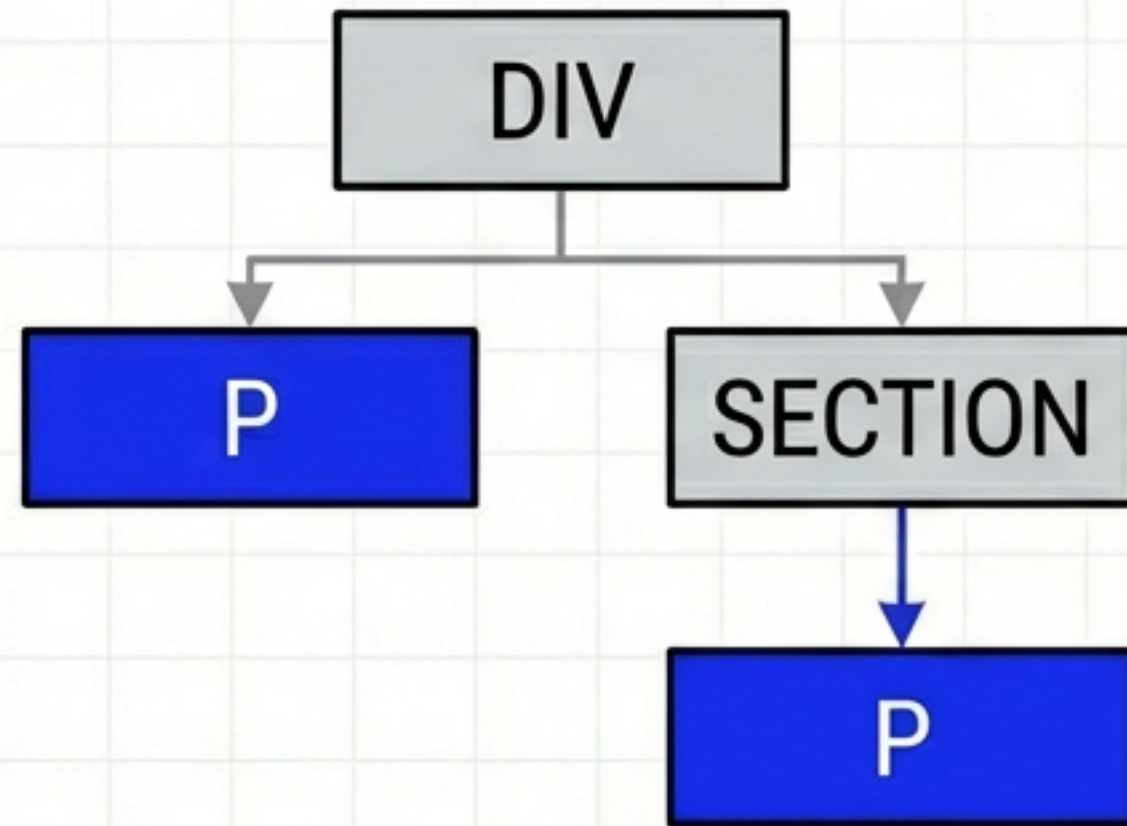
Pattern matching elements based on HTML attributes.

Pattern	Meaning	Code Example
[att]	 Has attribute	a[target] { color: pink; }
[att='val']	 = value Exact match	input[type='button'] { background: blue; }
[att^='val']	 ^value Starts with	a[href^='https'] { lock-icon: visible; }
[att\$='val']	 \$ Ends with	img[src\$='.png'] { border: 1px solid grey; }
[att*='val']	 * Contains	div[class*='menu'] { background: grey; }

Family Matters: Descendants vs. Children

Descendant Selector (Space)

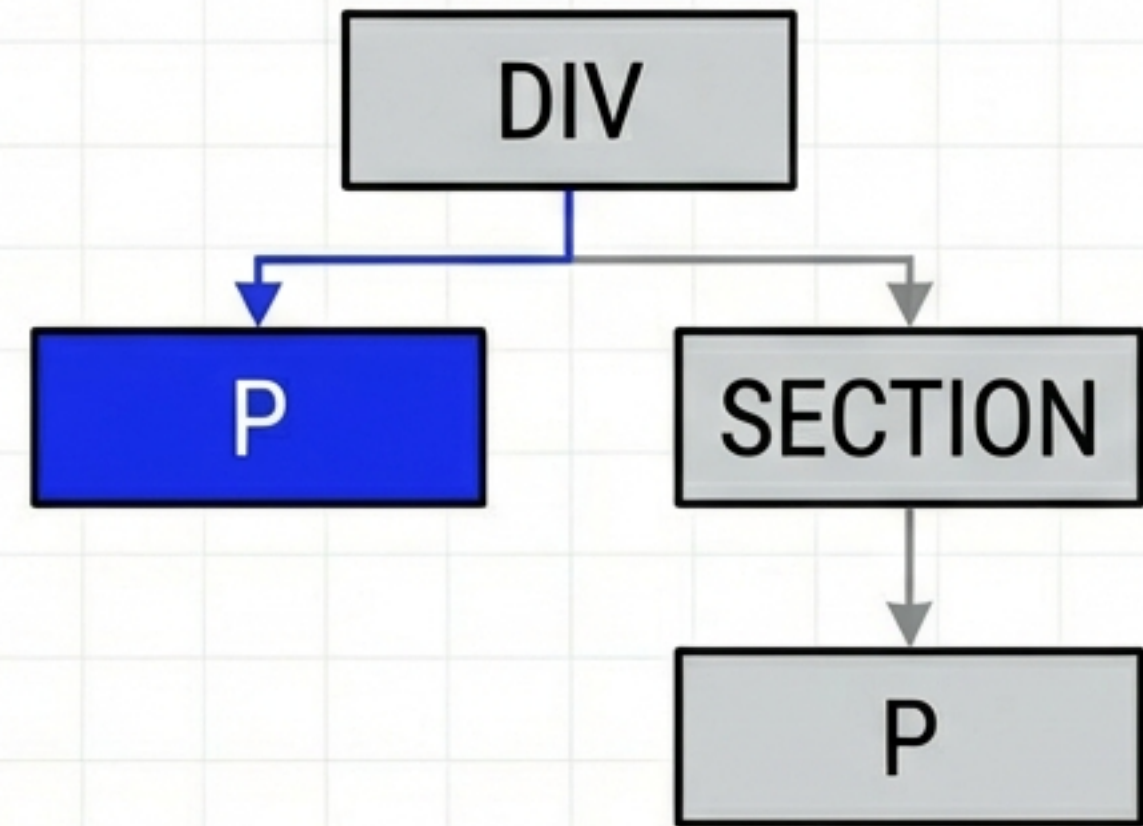
```
div p { color: blue; }
```



Matches ANY descendant,
no matter how deep.

Child Selector (>)

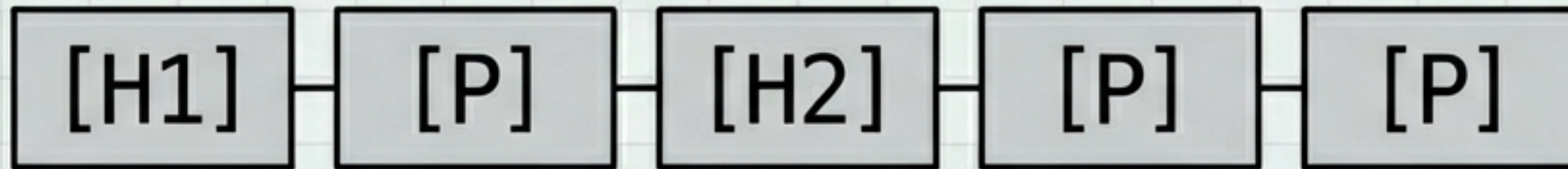
```
div > p { color: blue; }
```



Matches ONLY immediate
direct children.

Sibling Rivalry: Adjacent & General

Selecting elements that share the same parent.



Adjacent Sibling (+)

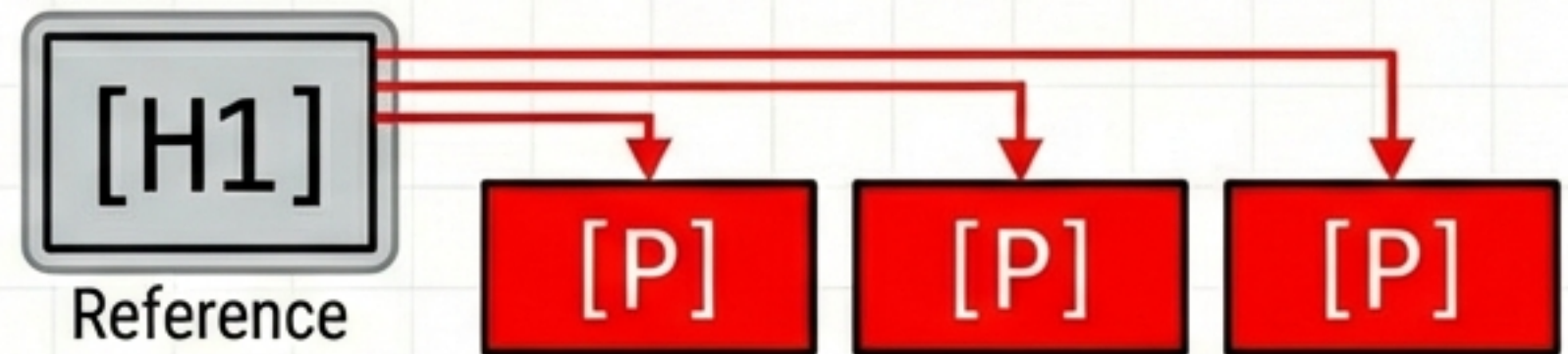
```
h1 + p { color: red; }
```



Matches the element immediately following.

General Sibling (~)

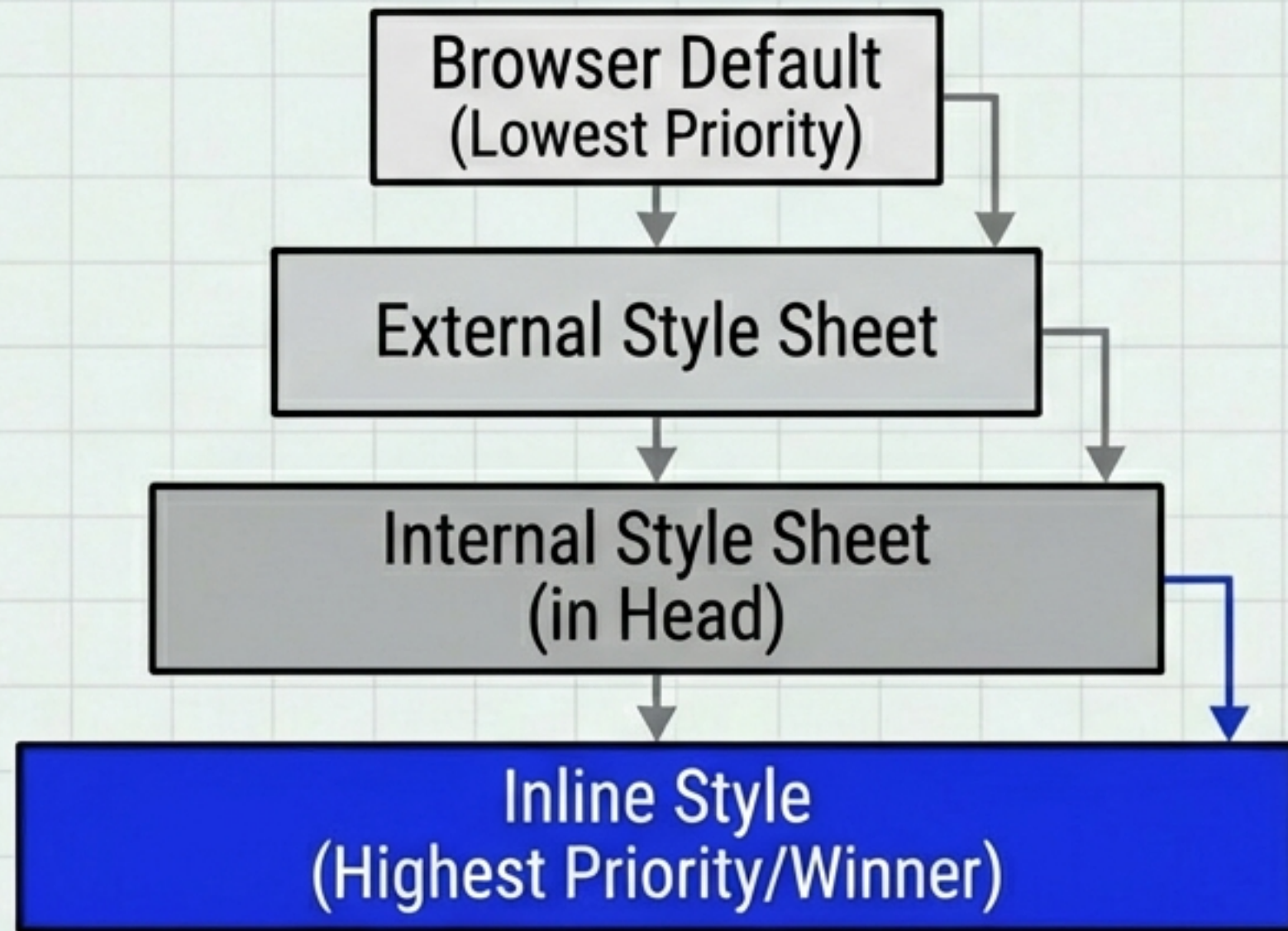
```
h1 ~ p { color: red; }
```



Matches all designated siblings following the element.

The Law: Cascade & Inheritance

The Cascading Order



Rule: Nearest Wins.

Inheritance

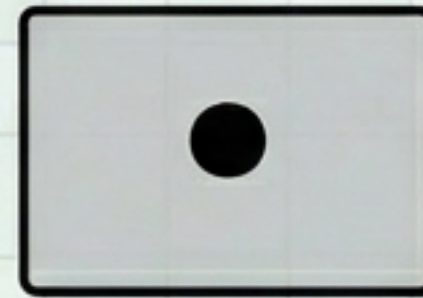
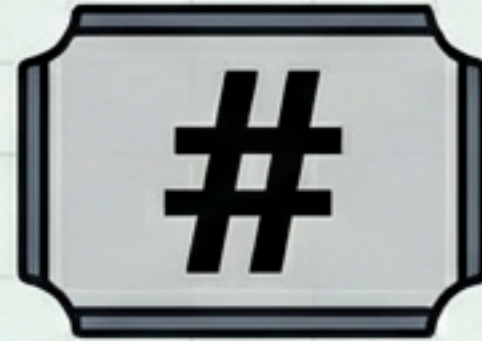


What gets passed down from parent to child?

- **Inherited:** Text properties (color, font-size, line-height, text-align).
- **NOT Inherited:** Box properties (width, border, margin, padding).
- **Exception:** Anchor tags `<a>` do not inherit color by default.

Specificity: The Scoring System

How the browser resolves conflicting rules.



Inline	ID	Class / Attribute	Element
1,0,0,0	0,1,0,0	0,0,1,0	0,0,0,1

The Universal Selector (*) has a score of 0,0,0,0.

Calculating Specificity Scores

Highest score wins. Ties are broken by the last rule defined.

Selector	Breakdown	Inline, ID, Class/Attr, Elem.
#warning	1 ID	0 + 1 + 0 + 0
p.message	1 Class + 1 Element	0 + 0 + 1 + 1
p#warning	1 ID + 1 Element	0 + 1 + 0 + 1
div p.message	1 Class + 2 Elements	0 + 0 + 1 + 2

← Highest
Score
Wins!

The Universal Selector (*) has a score of 0,0,0,0.

The Nuclear Option: !important

Function: Overrides all specificity and cascade rules.

Syntax:

property: value !important;

Example:

color: red !important;

Hierarchy of Power

1. !important



2. Inline Styles

3. ID Selectors

4. Class / Attribute Selectors

5. Element Selectors

6. Universal Selector (*)

With great power comes great responsibility. Use !important sparingly.